SHARNOUBy AI

Business Logic and Technical Report

Prepared for Arwa Saadawy | 12 October 2026

Executive summary

SHARNOUBy AI is a local business decision support application that connects recorded sales performance, product forecasts, inventory positions, production allocation, and financial scenarios. The application gives business users a consistent route from evidence to a reviewed planning decision, with reports and calculation exports that preserve the assumptions behind each result.

The application consolidates previously separate views into six workspaces and twelve views. Its default Live Business Report combines observed performance, conditional year end estimates, production inputs, business discoveries, anomaly evidence, and action follow up. Product analysis includes forecasts, while inventory analysis includes production and material planning.

The main findings in the company wide September 2026 scope are a 17.24% decline in net recorded sales versus August, a 5.05% increase in average order value, and a 20.94% decline in invoice count. Pajamas and homewear account for the largest negative category movement. Inventory evidence identifies both excess positions and replenishment needs, supporting review of transfers before additional production.

The selected demand model is Seasonal naive after comparison with six alternatives. Held out product forecast WAPE is 26.78%. Separate models cover stockout risk, repeat purchase, weekly pricing associations, product segmentation, and unusual days. The application exposes their different evaluation results rather than presenting a single general AI accuracy figure.

Scope and interpretation

Operating records in the supplied project are modeled training data. Findings in this report describe that dataset and demonstrate the application logic; they do not establish audited company performance. Sales records span 1 October 2023 to 30 September 2026. The default insight scope is all sales locations and all product categories during September 2026, compared with August 2026. Future forecasts originate at 30 September 2026.

The application supports analyst review and conditional planning. Realized financial improvement, production execution, transfer completion, and live provider connectivity require separate operational verification. The report separates implemented functionality, measured results, and proposed improvements.

Business objectives and data coverage

The business problem is the separation of commercial and operational decisions. A sales decline, a shortage forecast, or a profitable looking product can each lead to an incomplete decision when stock availability, returns, production capacity, and cost assumptions are considered separately. SHARNOUBy AI combines these perspectives within a shared business scope.

Decision questions

The application helps a user identify which categories explain recorded changes, which products merit investigation, where stock is insufficient or excessive, whether transfers can reduce local gaps, how limited capacity should be allocated, and what the resulting financial estimate means under explicit assumptions.

Data domain	Records	Primary purpose
Sales transactions	180,000	Invoices, units, prices, refunds, estimated cost
Product catalog	59	Product identity and family relationships
Customer master	30,000	Audience identity and consent
Inventory history	6,372	Monthly balances across product and sales location
Production orders	2,500	Completed output, yield and delay
Printing batches	1,900	Printing quantities and rejection
Purchase orders	3,600	Supplier receipt and acceptance
Quality inspections	9,000	Inspection evidence
Marketing campaigns	420	Spend and attributed revenue
Daily web traffic	1,096	Sessions and orders
Customer service tickets	6,500	Response time and satisfaction

The location master contains five locations. Three are sales locations used in the current 177 product and location inventory positions: the Sidi Bishr store, the online store, and the wholesale desk. Head office and the production hub are also listed, but they should not be interpreted as five independently modeled sales branches.

Additional inputs include business units, group operations, materials, suppliers, product bills of materials, machine downtime, case briefs, and a data dictionary. Machine downtime is retained in the data layer but has no dedicated predictive maintenance interface in the application. No image dataset or reconciled size and color inventory ledger is supplied.

Business rules and calculation definitions

Shared scope and comparison dates

Sales based views share location and product family filters. Historical periods control observed calculations. A full selected month is compared with the previous calendar month; a custom daily interval uses a preceding interval of equal length. A prior period outside available history produces an unavailable comparison. Annual closing and physical count use a year selector instead. Planning views use the latest completed inventory snapshot and fixed forecast origin.

Revenue and return recognition

Invoiced net sales are the sum of recorded Net_Sales_EGP for orders within the selected Order_Date interval. Refunds are recognized using Return_Date, including refunds for purchases made in earlier periods. Net recorded sales equal invoiced net sales less return date refunds. This convention prevents a later refund from being inserted into an earlier historical sales period.

Average order value equals invoiced net sales divided by distinct invoice count before event date refunds. In this dataset every invoice ID is unique to its sales record. The source does not establish multi product baskets, so catalog similarity cannot be described as a frequently bought together model.

Units represent sold quantity before returns. Returned unit quantities are not available in the sales ledger, so the system does not fabricate return adjusted unit counts. Gross profit and margin use estimated cost information and must be labeled accordingly.

Financial and inventory boundaries

Financial closing subtracts estimated sold cost of goods and documented expenses, and adds only manually documented recovery of returned goods cost. Purchase order costs are not deducted again on top of sold cost of goods. Production plan cost is a planning objective and is not automatically booked as an additional financial expense.

An inventory position is recommendation eligible only after stock arithmetic and the sales ledger comparison pass. Invalid positions remain visible for review, but they cannot receive automatic production allocations or act as transfer donors or recipients. Proposed transfers do not alter physical stock or become completed movements.

Output class	Meaning	Example
Fact	Recorded arithmetic or source observation	September net sales and current ledger mismatch
Prediction	Uncertain estimate evaluated where possible	Next month units or stockout probability
Recommendation	Decision conditional on explicit assumptions	Production quantity under entered capacity

Live business report and financial planning

The Live Business Report is the default business workspace. It assembles the same scope, financial rates, forecasts, inventory, and product evidence into one reproducible calculation package. It replaces the standalone overview and integrates discoveries and action review.

Inputs and outputs

Financial controls include future selling price change, future unit cost change, assumed demand fulfillment, a reviewed remaining expense budget, and a full year result target. Production controls include available capacity, reference unit production cost, shortage penalty, holding cost, and explicit approval to include proposed transfers. Production controls are shared with Inventory Command.

The report shows selected period net recorded sales, the conditional year end result, allocated production, financial scenario charts, historical KPI tables, annual year to date calculations, forecast months, break even calculations, margin watch, inventory positions, and production allocations. Business discoveries and unusual days are embedded alongside action follow up.

Conditional financial conversion

For each product, the financial layer converts forecast units into fulfilled units using the entered fulfillment fraction. Fulfilled units multiplied by historical net price and the price adjustment yield estimated invoiced sales. A historical refund event ratio estimates refunds. Estimated sold cost uses historical cost per unit and the cost adjustment. Contribution equals estimated net sales less estimated sold cost.

Historical price and cost rates use the past 90 days, then the past 365 days or all earlier sales when necessary. Refund rates use past 365 day event date refunds relative to invoiced sales. Invalid historical rates prevent unsupported profit estimates.

Full year result equals the recorded year to date estimated result plus forecast remaining contribution less the entered future expense budget. A missing budget is shown as missing; the displayed estimate remains before unknown future expenses. Annual forecast totals are withheld if supported forecasts do not cover the remaining source year.

Stress scenarios and target logic

The lower demand stress uses lower forecast units, an additional five percentage points of cost increase, and a refund rate multiplier of 1.25. The central case uses entered assumptions and central units. The upper demand case uses upper units. These are explicit stress cases, not a calibrated confidence interval for annual profit.

Break even net sales equal entered future expenses divided by the contribution ratio, when positive. Target gap compares the full year result with the user target; additional sales use the same contribution ratio and unchanged mix assumption. Downloads include readable HTML, full CSV calculation tables, and assumption metadata; HTML previews are limited to 200 rows per table.

Product intelligence forecasts and seasonality

Product intelligence

A searchable product selector connects period sales, units, change, estimated margin, category membership, descriptive segment, future forecasts, and current inventory. Users can inspect size preference within the selected sales period and review stock across locations. The selected product carries into scenario and inventory workflows.

Cosine similarity ranks products using catalog attributes. It is descriptive content similarity, with no claim of basket association or conversion likelihood. Product clustering is computed across company history and is not silently recomputed when a branch or category filter changes.

Forecasting inside the product view

The product view includes a historical and future unit chart, one to three future months, lower and upper bounds, model name, product validation WAPE, and a historical reliability category. It also displays company level candidate comparison metrics. Selecting a branch scales product forecasts by that branch share of recent company product units; it does not retrain a local model.

High, Medium, and Low labels summarize product development WAPE below 20%, below 40%, and otherwise. They are not probabilities that a prediction will be correct. Historical charts aggregate whole months overlapping the selected dates; edge months may include days outside the exact daily window.

Seasonal planning

High Season uses complete calendar years and computes each month units divided by the mean monthly units in that year. The seasonal index is averaged across complete years. With the default threshold of 1.2, a high season signal requires an index at least 20% above the same year average and at least two observed complete years.

Users select a target calendar month and preparation buffer. Product tables combine historical seasonal strength with forecasts when that target exists in the trained horizon. Prepare by dates subtract supplier lead time and the entered buffer from the target month start. The rule does not simulate factory queues, transit, or intervening demand.

A central seasonal stock gap is shown only when forecast coverage exists and all relevant product inventory positions are reconciled. Outside the trained horizon, historical target month averages remain descriptive and unsupported future quantities are withheld. Holiday dates and causal seasonal explanations are not inferred.

Business use

A buyer can move from a product with falling recorded volume to its next month forecast and inventory availability, then evaluate whether a supply action is warranted. A seasonal rise is reviewed separately from historical product momentum, preventing a predictable seasonal pattern from being presented as evidence of new sustained growth.

Inventory monitoring transfers and action follow up

Inventory Command works at product and sales location grain using the latest reported completed month balance. It combines stock, forecast allocation, supply lead time, uncertainty proxies, and a separately trained stockout classifier.

Coverage and replenishment logic

Daily demand is the allocated next month forecast divided by the number of calendar days in that forecast month. Coverage days divide closing stock by this daily demand. A safety buffer scales the difference between the upper and central forecast by the square root of lead days divided by forecast month days. Reorder point combines expected lead time demand and that buffer.

Suggested units cover central next month units plus the buffer above existing stock. Excess units are stock above the upper forecast plus buffer. Risk labels identify replenishment needs, excess, monitoring positions, and positions requiring data review. These are planning proxies rather than calibrated service level guarantees.

Transfer proposals

For each product, eligible surplus locations can supply eligible locations with a planning gap. Quantities are bounded by donor excess and recipient need. Unit conservation and donor protection are enforced. The current release proposes routes and quantities but does not verify transit times, route availability, minimum order quantities, or actual transfer costs.

Approving proposed transfers as a planning assumption changes available stock and net production need in the optimization calculation. It does not execute a movement. When a single branch is selected, a proposed alternative can involve other branches because the transfer is evaluated across the company product stock positions.

Action review

The Live Business Report ranks current inventory evidence and exposes the largest central forecast gaps. Each item includes stock, allocated demand, coverage, risk, classifier probability, and forecast reliability. Users acknowledge, dismiss, or reopen items; acknowledged and dismissed items can be included again for review.

Action IDs identify the product, location, and forecast month. Operator state is stored in data/workspace/actions.db, separately from model artifacts. Existing V5 action rows are imported on first use. The review status records an operator response and should not be interpreted as approval or completion of production or transport.

Revenue exposure is a ceiling obtained from the forecast gap and historical unit invoiced revenue. It is not measured lost sales and does not establish the revenue recoverable through replenishment. The stockout classifier probability and the inventory forecast gap answer different questions and are displayed separately.

Production optimization and material requirements

The production planner is embedded in Inventory Command and is also calculated in the Live Business Report. Its purpose is to allocate a limited monthly production budget across eligible products under explicit cost assumptions.

Inputs and formulation

Inputs are total available units of production capacity, a reference production cost, a shortage penalty, and an excess holding cost. Product unit production costs scale the reference input by historical product estimated cost per unit relative to the median valid product cost. This makes costs product specific without claiming an audited manufacturing cost model.

The linear programming model uses lower, central, and upper demand scenarios with weights 20%, 60%, and 20%. Decision variables cover production, shortage, and excess for each product and scenario. The objective minimizes production cost plus weighted shortage and holding costs. Scenario weights are planning assumptions, not estimated probabilities.

Total production cannot exceed entered capacity. Each product production is bounded by its eligible net recommended need. Available stock accounts for transfer inputs and outputs only when the proposed transfer assumption is approved. Invalid stock positions are excluded before aggregation.

After solving with SciPy linprog, production is rounded down to whole units and the cost components are recalculated from rounded quantities. This can leave a small unused fraction of capacity. No separate button is required to recompute the plan when its inputs change.

Outputs and material linkage

The planner exposes produced units, unmet recommended need, available stock, product production costs, weighted shortage, weighted excess, and total planning scenario cost. It also compares updated quantities with a prior plan within the same scope, indicating whether allocation changed or only cost values changed.

The Bill of Materials joins product production quantities to required material units. Required material equals quantity produced multiplied by required material per finished unit, aggregated by material and measurement unit. This output estimates consumption; it does not confirm current raw material availability or supplier commitments.

Worked default example

In the company wide current inventory scope, capacity of 5,000 units, reference unit cost of EGP 100, shortage penalty of EGP 200, holding cost of EGP 10, and unapproved transfers produce an allocation of 4,999 units. Recomputed production cost is approximately EGP 446,316; weighted shortage cost EGP 1,099,604; holding cost EGP 535,510; and total planning objective EGP 2,081,430. These values are scenario costs, not booked expenses.

Line level schedules, machine availability, batch changeovers, transport constraints, and raw material constraints are not part of this optimization model. The recommended quantities require operational review before becoming a production order.

Scenario testing and customer analytics

Weekly scenario testing

What if Lab compares the last modeled product baseline with an alternative complete week. Controls include supported list price, discount fraction, scenario opening stock, additional available units, and assumed cost per new unit. Branch results use recent company product sales shares. The interface shows the modeled week and keeps weekly output separate from monthly demand forecasts.

The histogram gradient boosting model estimates units using observed weekly price and discount associations. Price inputs stay within observed ranges; a constant observed list price is disabled for editing. Discounts outside observed support return no demand estimate. Small price or discount changes can leave a tree model prediction unchanged, while stock and revenue arithmetic still update.

Supply equals scenario opening stock plus additional available units. Central shortage is modeled demand above supply; excess is supply above modeled demand. Revenue ceiling before returns uses the lesser of modeled demand and supply multiplied by price after discount. Additional production cost uses entered new units and the unit cost assumption.

The output is a conditional observational scenario. It does not establish causal promotion uplift, an exact future stock balance, realized revenue, or a calibrated counterfactual stockout probability. Future intervening sales, returns, production delay, and transit are not simulated.

Customer audience definitions

Customers supports period purchasers or all known buyers up to the selected end date. Location and product category apply to historical customer evidence. Recency is time since last scoped purchase; frequency and monetary value use the prior 365 days. Recent frequent customers have recency at most 60 days and at least three orders; inactive customers have recency above 180 days. Other rule labels are Regular and Low frequency.

An opted in consent filter is enabled by default. Audience exports include scoped activity and available current repeat purchase scores. Customers without a current model score remain in the audience with an unavailable probability rather than receiving an invented value.

The repeat model estimates any company wide purchase within 90 days at the latest model origin. It is not specific to a selected category or a backdated forecast at the historical filter end. Session and date labels make this distinction explicit.

Business use

A user can build an eligible campaign audience from observed behavior, then use model probabilities as exploratory supporting evidence. Strong RFM engagement should not be confused with a validated marketing response model. The weekly scenario view helps bound a promotion experiment; an actual campaign still needs measurement and approval outside the application.

Operational performance reporting

Operations consolidates factory, support, procurement, and digital ledgers through a report selector. These source ledgers do not provide a sales branch key, so location filtering is disabled. Category filtering is applied only where a product relationship exists. Each report declares its event date basis.

Production printing and quality

Production uses completed orders whose Completion_Date lies in the selected period, reporting full completed order good units, delay frequency, yield, and line level orders, rejects, and average delay. Printing uses Print_Date to calculate rejection quantity divided by printed quantity. Quality inspection records use Inspection_Date and remain available as supporting evidence.

Procurement and marketing

Procurement selects receipts by Actual_Receipt_Date. Late days compare actual receipt with Promised_Date, clipped at zero. On time receipt share counts orders arriving on or before the promised date; accepted quantity divided by ordered quantity summarizes acceptance.

Marketing selects campaigns whose End_Date lies in the historical interval. Spend and attributed revenue remain whole campaign totals. Attributed revenue divided by spend is a descriptive ratio; it is not incremental return on investment. Campaign totals are not artificially prorated into daily amounts.

Digital support and group operations

Web funnel uses dated sessions and orders from its own ledger. Orders divided by sessions is a recorded conversion proxy. Separate charts prevent sessions and orders from being visually confused. Web orders are not silently reconciled to sales invoice counts.

Customer service selects tickets by Open_Date, then reports final recorded first response and satisfaction values. Missing resolution timestamps prevent reconstructing a historical as of ticket status. Group operations selects whole months overlapping the period and keeps business unit KPI units separate instead of summing unlike measurements.

September 2026 measure	Result	Interpretation
Completed good units	152,137	Output of 61 completed production orders
Production yield	97.97%	Good units divided by produced units
Delayed completed orders	8.20%	Five of 61 completed orders
Printing rejection fraction	2.51%	Selected print date batches
On time receipts	56.10%	46 of 82 received purchase orders
Marketing revenue to spend	7.50 times	Attributed revenue only
Web orders to sessions	2.85%	1,356 orders and 47,652 sessions
Mean first response	10.57 minutes	186 tickets opened
Mean satisfaction	3.87 out of 5	Recorded CSAT

Operational quantities and sales quantities have different grains and time bases. Production output should not be compared directly with September sales units as evidence of a stock buildup without reconciling movements and product coverage.

Annual closing and physical inventory counts

Annual financial reconciliation

Annual Closing selects a calendar year and the current business scope. The source covers complete 2024 and 2025 calendar years, a partial 2023 year, and 2026 through September. Incomplete years are labeled partial rather than treated as completed closes.

The workspace includes a yearly result, monthly revenue and refund reconciliation, invoice and refund registers, searchable invoice IDs, and a documented expense ledger. Invoice searches change the visible register but not the annual cards. Refunds recognize events during the selected year, including earlier year purchases.

Supported entries are operating expense, finance cost, tax expense, inventory write down, and return cost recovery. A date, amount, scope, and document reference are required. Duplicate active entries with matching date, type, scope, and reference are rejected. Reversing an entry retains its audit history. Unallocated company expenses are not silently assigned to a branch or category slice.

Estimated annual result equals invoiced net sales less event date refunds less estimated sold cost, plus documented return cost recovery and less entered expenses. It is not certified net profit. The ledger does not infer payment settlement, cash balances, receivables, automatic tax, or VAT separation.

Physical count workflow

Annual Count prepares book inventory at an available cutoff for the chosen year and scope. If a December closing balance is unavailable, the interface displays an interim cutoff. Counts made today cannot be compared as if they were historical counts at a different date.

Users download a blank count template, edit actual units, or import a completed CSV. Blank actual units mean not counted; zero means counted with no units present. The system validates product and location identities, rejects invalid entries, and computes difference as actual units less book units.

Counted positions report net difference, separate gross shortages and surpluses, and positions requiring reconciliation. Cost values use a historical estimated cost per unit proxy. Missing cost remains unavailable. Saving creates a separate snapshot with a required count reference; it does not overwrite book stock or automatically book an expense.

Decision control

A count discrepancy calls for investigation of receipts, sales, returns, damage, and counting errors. It is not automatically classified as fraud or lost revenue. A reviewed inventory write down can be recorded separately in Annual Closing, with care to avoid duplicate recognition.

Annual adjustment and count sessions remain in the existing business database. Moving installations or rebuilding models should preserve these records through the supplied migration and training preservation logic. Action history uses its separate workspace database.

AI copilot and evidence controlled conversation

AI Copilot provides an English only interface to business evidence. It can retrieve sales summaries, comparisons, product facts, forecasts, inventory priorities, customers, anomalies, seasonal evidence, invoices, annual calculations, physical count evidence, and supported year end financial outlooks.

Conversation workflow

Without provider configuration, deterministic intent routing and allowlisted evidence tools produce a guided business answer. With a configured provider, the model may route a question to a read only evidence function, after which local calculation tools retrieve the result. A further provider request can add a short qualitative explanation. Numerical evidence remains in the application tables.

Conversation stores recent messages and product and year context within the session. Changing branch, category, historical dates, source database modification time, or selected chat year starts a fresh context. Users can clear the conversation and export messages with evidence. Context persistence is session level rather than durable cross session memory.

Language and provider behavior

Arabic input is rejected in the interface. Response strings and object values are filtered for Arabic source text, and provider narrative is checked for English and qualitative content. Narrative containing digits or unsupported text is withheld while verified tables remain available. Numeric safety is a practical output check, not a formal proof against every possible unsupported qualitative claim.

Provider configuration requires an API key, model ID, and HTTPS Chat Completions base URL. Controls apply the connection for the session, test connectivity, or save configuration locally to .env. Configured conversation defaults to enabled. Missing credentials leave the offline evidence assistant usable.

Requests use a 30 second timeout. Authentication, permissions, model or endpoint, rate or quota, invalid request, and network failures receive actionable messages without echoing credentials. Saved .env credentials are local plaintext configuration and require local access protection. Downloaded reports do not include the key.

Current financial context boundary

The generic chat year end tool invokes the financial outlook without the reviewed future expense budget or the current report financial sliders. Its output therefore explicitly warns about missing future expenses. Users must review scenario specific results in Live Business Report; chat should not be described as automatically reproducing every entered report scenario.

Verified status

LLM request handling, routing structure, fallback behavior, and redacted errors are implemented. No API key is bundled and no live provider connection was verified during release validation. The application does not autonomously execute production, transfer, procurement, campaign, or accounting actions.

Verified sales insights and business interpretation

All figures on this page use the company wide all category scope, 1 September to 30 September 2026, compared with 1 August to 31 August 2026. Monetary values are EGP. Calculations were recomputed from the delivered data.

Measure	August 2026	September 2026	Change
Invoiced net sales	13,312,040.63	11,055,736.59	-16.95%
Return date refunds	311,613.33	296,780.74	-4.76%
Net recorded sales	13,000,427.30	10,758,955.85	-17.24%
Sold units before returns	54,091	44,198	-18.29%
Distinct invoices	6,408	5,066	-20.94%
Average order value	2,077.41	2,182.34	+5.05%

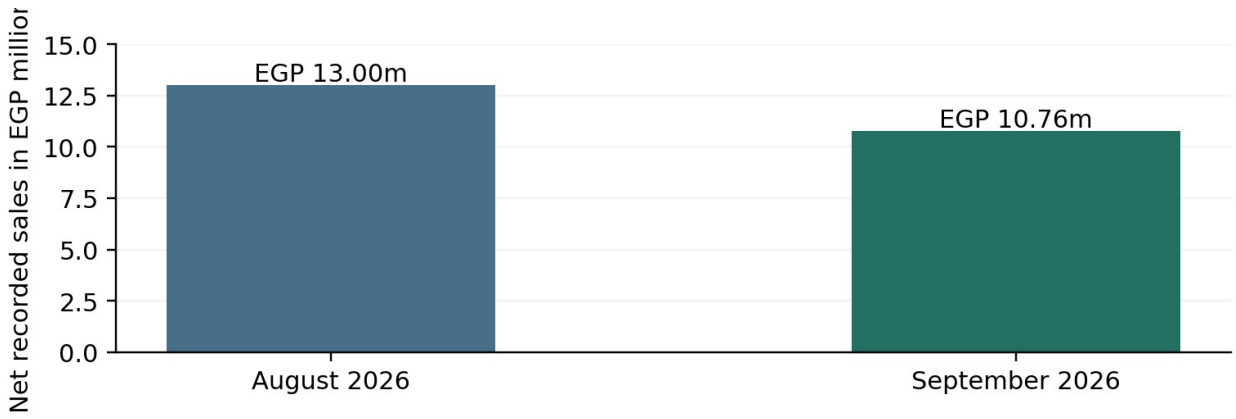


Figure 1 Recorded sales after return date refunds

The EGP 2,241,471.45 net sales decline occurred alongside fewer invoices and fewer sold units, while average invoice value increased. This describes an arithmetic pattern: transaction volume weakened while average order value rose. It does not establish whether acquisition, availability, price, or seasonality caused the decline.

Product family	September net sales	Change versus August	Growth
Kids Sets	1,231,118.00	-360,347.00	-22.64%
Pajamas and Homewear	7,163,816.01	-2,405,439.97	-25.14%
Sweatshirts	1,231,645.82	+758,510.76	+160.32%
T Shirts	1,132,376.02	-234,195.24	-17.14%

Pajamas and homewear account for the largest negative contribution. Their decline exceeds the total decline because sweatshirt gains partly offset other reductions. Category changes reconcile exactly to the overall movement. The business response is to investigate product and channel mix and availability, while reviewing seasonal sweatshirt replenishment rather than treating all categories identically.

Verified product inventory and seasonal insights

Product momentum and concentration

P001 recorded September net sales of EGP 546,109.44 and 2,379 units, the highest product net sales in the selected period. P008 and P051 followed at EGP 509,560.74 and EGP 508,360.51. High current revenue is distinct from growth: P001 units declined 10.97% versus August.

P032, the blue boys sweatshirt, increased from 209 to 816 units, a gain of 607 units. P045 increased from 199 to 663, P038 from 295 to 744, and P042 from 67 to 506. These are historical momentum observations. Availability and supported future forecasts still need review before additional production.

The wholesale desk accounts for EGP 9,133,985.37 of September net recorded sales, approximately 84.90% of the total. The online store accounts for EGP 956,643.12 and Sidi Bishr for EGP 668,327.36. This concentration supports separate monitoring of wholesale exposure; it is a dataset concentration measure, not proof of a business dependency outside the source.

Inventory imbalance

Current status	Positions	Business response
Excess	108	Review transferable surplus and supported stock scenarios
Replenish	33	Inspect forecast gaps and supply options
Monitor	29	Maintain observation
Review data	7	Reconcile balances before automatic planning

The current plan produces 65 proposed transfer records totaling 7,387 units. They are alternatives for review rather than executed transfers. P033 at the wholesale desk has reported stock of 45 units against an allocated next month forecast of approximately 1,958.6 units. Its central gap is approximately 1,913.6 units and its unapproved transfer planning need is 2,073 units including the buffer.

P043 and P039 at the wholesale desk also have large central gaps, approximately 1,552.6 and 1,492.7 units. These rows support prioritizing review of replenishment, lead time, and eligible stock alternatives. They do not prove an exact number of lost sales.

Seasonal and anomaly evidence

Complete years 2024 and 2025 support the seasonal calculation. December seasonal index is 1.2433, approximately 24.33% above the average month and above the default 1.2 threshold. July is 1.1815 and August 1.1241, below that threshold. The latest forecasts total 37,453 units for October, 42,307 for November, and 58,762 for December 2026.

The company wide September anomaly scope contains 30 evaluated days and zero flagged days. This is an explicit statistical result; it does not imply that the sales decline or inventory imbalance is absent.

Financial scenario results and commercial implications

These values use all company product forecasts from the 30 September 2026 origin, unchanged price and cost assumptions, 100% assumed fulfillment, and no entered remaining expense budget. The annual source year is 2026. The figures are financial estimates derived from forecast units and historical rates.

Scenario	Remaining net sales	Remaining contribution	Year end estimate
Lower demand and cost stress	22,055,392.81	2,265,264.63	16,873,031.04
Central assumptions	36,448,640.78	5,363,830.48	19,971,596.89
Upper demand stress	53,088,022.80	7,751,373.95	22,359,140.36

The year to date estimated result is EGP 14,607,766.41 before missing recorded expenses. In the central remaining period case, estimated sold cost is EGP 31,084,810.31 against net sales of EGP 36,448,640.78. The resulting contribution is EGP 5,363,830.48. The displayed EGP 19,971,596.89 year end amount is before unknown future expenses and cannot be presented as expected certified net profit.

Sensitivity to entered expenses

A hypothetical reviewed remaining expense budget of EGP 1,000,000 would reduce the central year end estimate by exactly that amount to EGP 18,971,596.89, assuming all other inputs remain unchanged. Increasing the budget to EGP 2,000,000 reduces it by another EGP 1,000,000. This is deterministic scenario arithmetic rather than a model improvement.

The central contribution ratio is approximately 14.72%. Under the same unchanged product mix, price, and cost assumptions, an EGP 1,000,000 remaining expense budget implies approximately EGP 6.80 million of remaining net sales to cover that budget. These break even values are conditional and should be recalculated using reviewed inputs.

Margin watch

Two products have negative projected contribution under the default assumptions. P013, the beige boys summer T shirt, has projected contribution approximately EGP -7,517.74 and a ratio of -2.96%. P048, the Semon children set, has contribution approximately EGP -725.89 and a ratio of -0.15%. Historical costs and refund assumptions should be reviewed before a pricing or discontinuation decision.

Recommended sequence

First verify stock and historical rate validity. Then review existing transferable stock, supported demand forecasts, and production constraints. Enter the expense budget and achievable fulfillment assumption before using a year end target. Finally, inspect product contribution rather than optimizing revenue alone. Realized outcomes should be measured after operational actions, using reconciled sales, stock, and cost records.

Financial scenario totals combine marginal product demand bands. Summed lower and upper values are stress assumptions, not calibrated portfolio bounds. The report does not estimate tax liabilities, cash collection, or financing needs from absent source fields.

Technical architecture and data engineering

SHARNOUBy AI is a Python application with a Streamlit interface, a processed SQLite database, offline training modules, serialized model artifacts, a registry, and export utilities. The system is intended for a local analyst workflow. Its architecture separates model building from interactive calculations.

Layer	Main components	Responsibility
Source and validation	data/raw and src/validation.py	Read CSVs, validate values and date provenance
Offline artifacts	src/engine.py and src/postprocess.py	Train models, create processed tables and registry
Shared business calculations	src/analytics.py and src/annual.py	Scope, dates, revenue, refunds and reconciliation
Decision calculations	src/inventory.py and src/profit_forecast.py	Production allocation and financial conversion
Report assembly	src/live_report.py and app/reporting.py	Current frames, assumptions and downloads
Application	app/Home.py and app/workspaces.py	Navigation, widgets, charts and session state
Conversation	src/chat.py src/copilot.py src/llm_connection.py	Intent routing, evidence retrieval and provider handling
Operator state	src/action_store.py and workspace databases	Action status, annual entries and count snapshots

Pipeline and grain safety

Raw CSVs are validated before being written to processed tables and the SQLite database. Product identifiers connect catalog, sales, forecasts, and material records. Location identifiers connect sales and inventory. Monthly demand panels use product and month grain, inventory uses product and location at a snapshot month, and weekly scenarios use product and complete calendar week.

Calculations aggregate each source at its intended grain before comparisons. They do not join all ledgers into a single row set and sum duplicated financial values. Product and month duplicates in financial forecasts are rejected to prevent double counting. Invoice and refund registers reconcile separately because their dates can differ.

Runtime and state

Streamlit caches loaded database tables and registry data using modification stamps. Interactive scope and assumption changes recalculate derived data without retraining models. Session state retains product selection, planning inputs, and conversation context. SQLite stores annual adjustments and physical counts; The application separates operator action state into its own small database.

Plotly renders interactive charts. Pandas and NumPy provide aggregation and arithmetic; Scikit learn supplies several model implementations; SciPy solves the production linear program. Statsmodels and XGBoost are candidates in the offline forecasting pipeline. The registry records model choice, feature definitions, evaluation periods, metrics, and raw source hashes.

Machine learning methods and measured evaluation

Task	Method	Measured result	Business interpretation
Monthly product units	Selected Seasonal naive	Test WAPE 26.78%	Uncertain monthly units, not unconstrained demand
Next month stockout	Random Forest with sigmoid calibration	Recall 88.89%; precision 35.82%	Sensitive alerts with substantial false positives
Repeat purchase within 90 days	Random Forest	ROC AUC 0.569	Exploratory ranking with modest discrimination
Product segmentation	KMeans with two clusters	Silhouette 0.530	Descriptive product grouping
Unusual recorded days	Isolation Forest	20 scope models	Unsupervised flags without labeled accuracy
Complete week product units	Histogram gradient boosting	WAPE 35.46%	Observed pricing associations
Financial contribution proxy	Forecast units and frozen rates	WAPE 3.43% over three months	Limited aggregate contribution backtest

Monthly demand validation

Seven candidates are evaluated: naive, Seasonal naive, moving average, exponential smoothing, Random Forest, XGBoost, and histogram boosting. Three expanding development folds use recursive three month forecasts. The final three months are excluded from candidate selection, followed by production refit through September 2026.

Seasonal naive uses product units from the same calendar month one year earlier. Its held out MAE is 238.81 units, RMSE 353.17 units, WAPE 26.78%, and aggregate bias -3.97%. Empirical absolute development residuals define horizon specific nominal 90% marginal bounds. Actual held out coverage is 82.49%, so nominal labeling does not establish achieved 90% coverage.

Risk and customer models

Stockout features include closing stock, sales rate, lead days, sold and damaged quantities, lagged stockout, month, and coverage. Separate chronological training, calibration, and test periods support probability calibration and a 0.06 threshold selected for a recall oriented objective. The test has 27 positives in 531 rows; precision of 35.82% means alerts need review despite ROC AUC of 0.965.

Repeat purchase uses past recency, frequency, monetary value, units, and average order value. Test recall is 14.49% and precision 93.44% at threshold 0.5, with ROC AUC 0.569. Repeated customers occur across chronological snapshots, so evaluation measures temporal generalization rather than performance on entirely unseen customers.

Weekly and financial validation

Weekly model WAPE improves from a 56.57% baseline to 35.46%, with measured marginal band coverage of 87.82%. It uses recorded sale weeks and observed discounts. The contribution backtest freezes rates before held out demand, producing MAE of EGP 57,192.92 across three aggregate months. Aggregation can cancel product errors; the 3.43% contribution WAPE is not evidence of validated net profit accuracy.

Data quality safeguards and technical limitations

Date provenance

The current audit identifies 171,457 missing Return_Date cells, all classified as not applicable for nonreturned sales. It records no invalid dates or unresolved applicable dates in the audited date columns. Dates are recovered only through exact agreement for a unique invoice where that relationship exists. Arbitrary forward fill, random dates, or dates invented from identifiers are not used.

Inventory reconciliation

The system checks closing stock against opening stock plus receipts less sales plus returned quantities less damage. It separately compares monthly inventory sold quantities with sales ledger units at the same product and location grain. Seven current positions fail at least one check and are blocked from recommendation eligibility. Source balances remain visible rather than being silently corrected.

Temporal leakage prevention

Demand features use lagged observations. Forecast selection uses chronological development and test periods rather than a random split. Stockout targets refer to the next month, with current completed month evidence as input. Customer features stop at each cutoff and target the following 90 days. Financial conversion freezes historical rates at the forecast origin; future price, cost, and return events cannot enter historical rate estimates.

Anomaly models use an early training segment and label later flags as prospective and early flags as retrospective. Classification feature importance is global impurity importance, not record specific SHAP evidence or causal effect. A descriptive distribution shift does not establish post deployment prediction degradation without actual future outcomes.

Security and operational boundary

The provider interface accepts HTTPS endpoints and uses redacted request errors. User and retrieved evidence strings are treated as untrusted data in the narrative prompt. The LLM has no arbitrary SQL or write tool. Provider endpoints and credentials must be reviewed before configuration, because retrieved bounded business evidence is sent to the configured service.

The local release does not include multi user authentication, role based authorization, production secret management, job scheduling, an enterprise backup policy, or an automated transaction approval engine. Pickled model files are suitable only when obtained from trusted project artifacts. Shared or public hosting requires additional engineering and access control.

Interpretation limits

Forecasts describe observed sales rather than demand that would exist with unlimited stock. Branch forecasts and uncertainty bounds are allocation proxies. Financial estimates use incomplete expenses and estimated cost. Promotions are observational scenarios. Physical counts need matching cutoffs. These limits belong to the calculation definitions and should remain visible in user decisions and external project descriptions.

Validation deployment and development priorities

Verification completed

Release checks exercised the twelve application views, customer audience filters, report recalculation, CSV and HTML export consistency, provider error redaction, accounting date rules, planning costs, English input behavior, product navigation, and durable action follow up. Fourteen targeted checks passed in the initial regression group. Later checks covered the action store correction and final live report path.

Tests support the implemented calculations and interface paths exercised; they do not validate business impact, every possible data condition, or a live LLM provider account. Provider connectivity requires credentials and remains unverified in the supplied release. No production model retraining was required for this report.

Local deployment

The Windows launcher creates or uses a Python 3.12 environment, installs requirements when imports are missing, and starts Streamlit at app/Home.py. Bundled processed data and models support first use. train_models.bat rebuilds artifacts after raw data changes; editing a raw CSV does not automatically retrain a cached forecast.

Model rebuilding stages a replacement database and preserves supported workspace records. Existing annual adjustments and count snapshots can be migrated using the supplied workspace migration module. Action history is stored separately and should be preserved when moving the installation. Local provider settings reside in .env and are excluded from project exports.

Recommended next development work

Priority	Proposed improvement	Reason
1	Reconcile real stock and reviewed cost records	Improve the evidence used in financial and supply decisions
2	Pass report assumptions into chat evidence	Make scenario specific copilot answers match current controls
3	Track forecast outcomes after deployment	Measure future error and calibration using actuals
4	Add material and line capacity constraints	Increase operational feasibility of production plans
5	Validate campaigns with controlled experiments	Measure causal promotion response and realized value
6	Add secure shared deployment and backups	Support multiple users and protect sensitive records

Success measurement should compare reconciled stockout frequency, excess stock, transfer completion, forecast error, decision turnaround, and realized contribution before and after use. Such measurement is proposed; the current dataset does not demonstrate a realized reduction in cost or an increase in revenue.

Source traceability and terminology

The numerical evidence and implementation descriptions are grounded in the delivered SHARNOUBy AI project. Historical guides and the inherited registry release string refer to prior releases in places; current navigation and the live report import path define the interface described here.

Evidence area	Project source
Data scope and recorded insights	data/processed/business.db and data/raw CSV files
Model identity and test results	models/registry.json and processed forecast validation tables
Scope revenue refunds and inventory rules	src/analytics.py
Annual entries and physical counts	src/annual.py and app/workspaces.py
Demand model selection	src/forecasting.py
Stockout and repeat models	src/risk.py
Segments similarity and pricing	src/intelligence.py and src/postprocess.py
Production cost optimization	src/inventory.py
Seasonality and prepare by rules	src/seasonality.py
Conditional financial calculations	src/profit_forecast.py
Current report exports	src/live_report.py and app/reporting.py
Conversation and provider interface	src/chat.py src/copilot.py src/llm_connection.py
Action state and connected navigation	src/action_store.py src/navigation.py app/Home.py

Working terminology

AOV is average order value. COGS is cost of goods sold, estimated here. RFM describes recency, frequency, and monetary purchase behavior. BOM lists required materials per finished unit. WAPE is the sum of absolute prediction errors divided by the sum of absolute actual values. MAE is mean absolute error; RMSE is root mean squared error. ROC AUC measures classification ranking across thresholds; precision and recall describe behavior at a chosen threshold.

Coverage is the share of measured outcomes falling inside a stated interval. Forecast horizon is how far into the future a forecast extends. Forecast origin is the last available training date. Contribution is estimated net sales less estimated sold cost before further future expenses. A planning assumption is an input selected for a scenario and is not itself a recorded fact.

Report reading guide

Business readers can use the executive summary, business rules, feature descriptions, and verified insights to assess decisions. Technical readers can use the architecture, evaluation, safeguards, and validation sections to assess implementation and evidence quality. Forecast and financial values must be read with their scope, dates, assumptions, and validation limits.